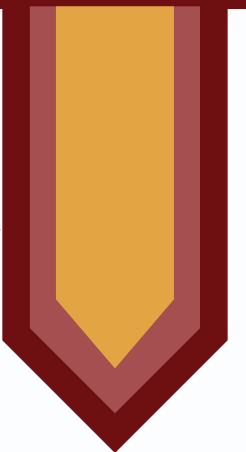


शुभ T2U

CASE 1 – ROUND 1

VOLT FLEET

IIM UDAIPUR
TEAM Ashwamedha



Raw Materials & Technology Sourcing

Manufacturing & Infrastructure

Fleet Management & Logistics

Regulation & Customer Interactions



1. Raw Materials Supply

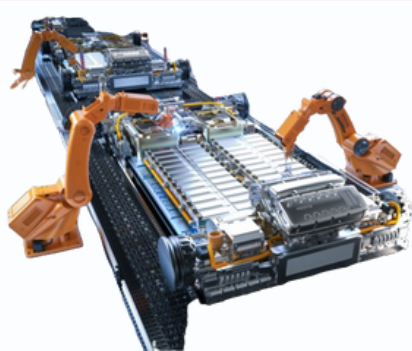
Supply of raw materials for LIB (Lithium Ion Battery) is forecasted to match the demand, if not exceeding. For example, Demand for Cobalt, one of the key materials in Battery Production, is forecasted to grow at a CAGR of 5.56% from 2021 to 2040, per the Elsevier Journal on Science Direct research.

We believe the supply would be enough to cater to this rising demand. For the last 20 years global cobalt production has grown with a CAGR of 6.77%. We expect it to continue because, as of 2023, the global production of cobalt reached an estimated 230,000 metric tons, and the total known reserves of cobalt worldwide are estimated at around 11 million metric tons . The same analysis has been done for Lithium and Nickel, and found the similar pattern.

2. Innovation in Battery

For the battery cathode materials, the industry is expected to move from cobalt-rich towards nickel-rich and Fe- and Mn-based since they are available in abundance. There are novel anode materials featuring silicon that replace or blend with graphite and improve energy density. Along with this, ongoing research is exploring alternative battery chemistries, including sodium-ion, iron-air, and liquid metal batteries which shows the role of batteries will continue to evolve.

Battery Price is expected to be cut down by huge margins in the next 5-10 years.



1. Indian Manufacturing Capability

The Economic Survey of India 2023 had forecast a robust 49 percent compound annual growth rate (CAGR) in India’s domestic electric vehicle market between 2022 and 2030, with an estimated 10 million annual sales by 2030.

To achieve the Global EV30@30 mission, the Indian government has highly supported Indian and Foreign EV manufacturers. 100% FDI approval will bring better technology to the country, and the knowledge will be shared with the domestic manufacturers.

Many Indian players have successfully ventured into this segment, like E-Trio, Piaggio & C. S.p.A, and Omega Seiki Mobility, and we expect the same to happen since the EV penetration in India is less than 2%. There is a huge growth potential in the market. More competition will lead to competitive pricing.



2. Charging Infrastructure

The EV-to-public-charging ratio in India is 135:1 when the global average is 6:20. This gap is due to factors like an inadequate power grid, the country's vastness, and the high investment cost. 5,254 public charging stations are currently operational in India. To meet its 2030 goals, India needs a total of 46,397 stations. This seems too difficult to achieve as per our analysis.



1. Fleet Management Solutions

The Fleet Management Solutions market is projected to grow at a CAGR of 15.25% from FY2025 to FY2032, reaching a market size of approximately USD 4.79 billion by FY2032. Integrating advanced technologies such as AI, GPS tracking, and telematics in fleet management solutions is expected to drive the market. These technologies enhance real-time tracking, fuel monitoring, driver management, and vehicle health monitoring, making fleet operations more efficient.

How Fleet Management Process Works



2. Indian Logistics Sector

The World Bank’s report highlights that logistics costs in India are among the highest globally, accounting for nearly 14% of the country’s GDP, significantly surpassing the global average of around 8%. These exorbitant costs not only hinder trade and competitiveness but also burden. Moving toward EV solutions is the way forward, as EV offers low TCO.

The Indian e-commerce market is projected to grow at 18-20% CAGR over the next decade. This will result in the growth of the Last Mile EV. Companies will move towards sustainable initiatives driving the growth in the logistics.



1. Indian Government Support

Electric Mobility Promotion Scheme 2024 (EMPS 2024) with a budgetary outlay of Rs 500 crore. Over four months, this scheme will support around 400,000 electric two-wheelers (e2W) and three-wheelers (e3W). The scheme is scheduled to end on July 2031.

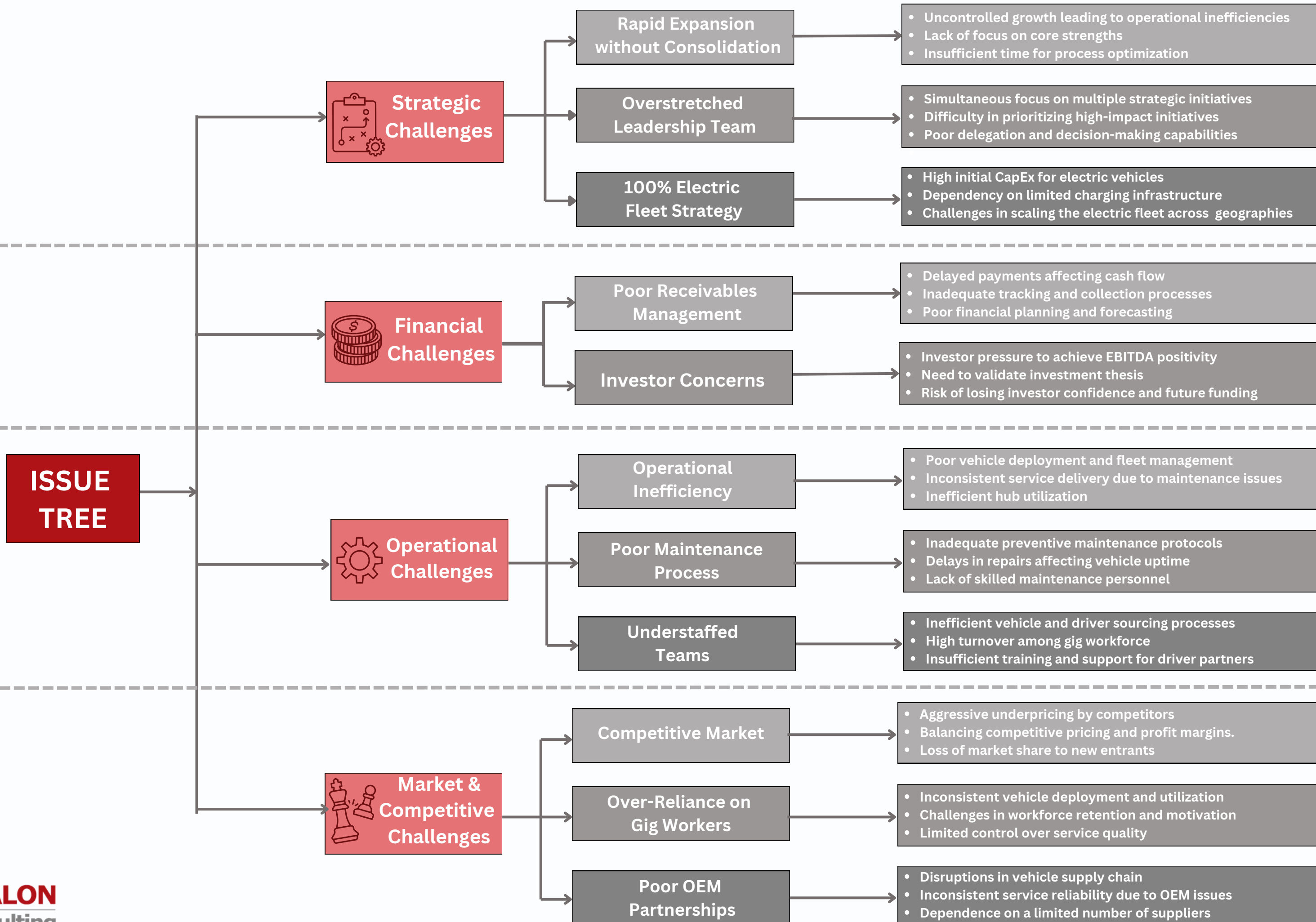
The government is expected to announce the FAME 3 scheme with an outlay of ₹10,000 crore. We expect these subsidies to continue till 2030 by the government to achieve its EV30@30 mission.

The government has put EVs under the 5% GST bracket. We believe this will continue until 2030 since the government is in talks to bring Hybrid vehicles under the same bracket. EV has high penetration in two-wheelers; they are likely to discontinue for 2 wheelers, but it will remain for others. The government will continue providing PLI Schemes to EV vehicle manufacturing and Charging station manufacturing.

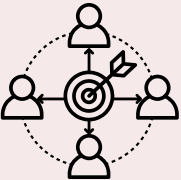
2. Awareness On Sustainability

India’s logistics sector has begun embracing EVs and adopting it faster than expected. For instance, Flipkart, Swiggy, and Myntra partnered with Zypp Electric to quickly transition to EV. Not only did Zypp help save 9 million KGS on CO2 through 2 million deliveries so far, but the company also enabled its partners in the grocery segment to cut down delivery costs by as much as 10%.

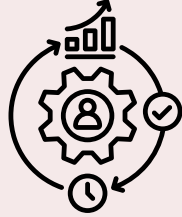




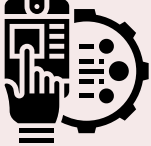
Hypothesis addressing the Strategic Challenges

Strategic Response	Implementation	Example/ Case Study	Impact	Effort	Priority
Consolidating Growth 	- Focus on Core business and process optimization - Reduce investments in Low impact projects - Consolidating Leadership and decentralized decision making	- Uber's Consolidation strategy post 2017, by exiting unprofitable ventures. - Apple, under Steve Jobs, streamlined product lines to focus on fewer, high-impact products, leading to iconic innovations and a company turnaround.	High	Low	V.High
Diversifying Fleet 	- Hybrid fleet strategy with non-EV vehicles in underdeveloped areas with poor Infrastructure to reduce CAPEX. - Better optimized blend of 2EV/3EV/4EV ownership model based on demand and geographic requirement.	FedEx uses a mix of electric, hybrid, and fuel-powered vehicles to optimize operations across various regions while lowering emissions.	High	Medium	High
Forge strategic collaboration 	- Collaboration with platform E-commerce players like Amazon/Flipkart/ONDC and to provide them 3-PL services. - Strategic coalition with other Mid-size logistic company by providing them appropriate technology, whereas utilizing their fleet for scaling of operations in new geographies	3PL Logistic firms have joined ONDC like Swiggy, Zomato, Amazon. - UPS partnered with Ware2Go, offering tech support in exchange for access to Ware2Go's warehousing network, enhancing logistics flexibility.	High	Low	Medium

Hypothesis addressing the Operational Challenges

Strategic Response	Implementation	Example/ Case Study	Impact	Effort	Priority
Centralized Fleet Management System 	- Improve deployment accuracy and maintenance of contractors - Preventive maintenance protocols to Minimize downtime and increase reliability - Adopt AI for route & load optimization to Reduce fuel consumption and delivery times	- UPS's ORION system optimized routes, reducing costs. - Delta's preventive maintenance reduced unscheduled repairs. - DHL's AI-driven route planning improved performance.	High	High	High
Workforce Scaling and fleet development 	- Aggregator Platform: Onboard gig workers via platforms like ONDC for flexible staffing. - Develop Own Fleet: Gradually build a self-owned fleet to improve service quality and competitiveness. - Smart Scheduling: Use demand-based scheduling to maximize workforce utilization throughout the day.	- Walmart blends full-time logistics staff with gig workers for consistent operations and to handle seasonal spikes. - Uber and Lyft use dynamic scheduling and geolocation to ensure drivers are available where demand is highest.	High	High	Medium
OEM Diversification & maintenance standardisation 	- Strengthen OEM partnerships for Ensure reliable vehicle supply and continuity - Standardize fleet model for reducing Maintenance cost by taking benefit of Economies of Scale	- DHL diversified suppliers to mitigate risks. - Airlines have standardized fleet to reduce the cost of spare parts and maintenance.	High	High	Medium

Hypothesis addressing the Financial Challenges

Strategic Response	Implementation	Example/ Case Study	Impact	Effort	Priority
Receivables Management System 	- Centralized A/R Dashboard: Integrate ERP for real-time visibility into A/R data. - Automated Collections: Streamline actions based on invoice aging and due dates. - AI-Driven Prioritization: Focus on high-value/risk accounts for optimized collections. - Unified Customer Interaction Logs: Record all customer communications for efficient dispute resolution. - Customizable Strategies: Tailor collections processes like escalation and communication. Unique payment plans for quicker payments from clients - Securitization of receipts for better capitalisation	- ZOHO and Oracle are the world leading companies in providing AI enabled smart receivable management system - For large receipts, Company can securitize it with bank and borrow money to ensure cashflow for operations. Example - E-NWR	High	Medium	V.High
Investor Communication 	- Transparent and open communication with investors to reduce anxiety and continued access to fund for growth - Strategic investor sourcing with investment in supply chain solution companies. This will help in access to resources other than capital	Tesla maintained consistent communication to its investors during production challenges. This helped them keeping investors onboard during difficult phase	High	Low	V.High
Fixed and variable cost cutting 	- Opt for vehicle leasing to reduce upfront costs and enhance fleet flexibility. - Centralize operations into fewer hubs to cut down on rent, utilities, and overhead. - Share warehousing with partners to reduce facility cost - Buy fuel and supplies in bulk to secure volume discounts. - Outsourcing Non-Core Functions: Delegate tasks like maintenance to third parties to lower operational costs	- FedEx saved costs by closing smaller facilities and focusing on larger hubs. - P&G and Unilever reduced costs by sharing warehouse space. - Amazon uses its scale to get discounts on fuel and supplies. - Outsourcing Non-Core Functions: XPO Logistics outsourced IT to cut costs while keeping quality.	V.high	High	V.High

Hypothesis addressing the Market & Other Challenges

Strategic Response	Implementation	Example/ Case Study	Impact	Effort	Priority
Differentiation and Premiumization 	- Service Customization: Offer tailored services like express or temperature-controlled delivery. - Bundling Services: Combine warehousing and delivery for a comprehensive package. - Premium Support & Reliability: Provide guaranteed delivery times and insurance for peace of mind. - Sustainability initiatives to become more attractive to conscious user	- DHL charges more for time-definite, climate-controlled shipping. - Amazon's FBA integrates storage, shipping, and customer service at premium rates. - FedEx's "Priority Overnight" service commands higher prices with a money-back guarantee.	V.High	Medium	V.High
Customer Centric Approach 	- Personalized Delivery: Using AI and data to tailor delivery experiences to customer preferences - Customizable Preferences: Allowing customers to set specific delivery options like time windows and drop locations - Hyperlocal Fulfillment: Partnering with local stores for ultra-fast, same-day deliveries - Utilizing crowdsourced drivers for faster, flexible deliveries	- Uber Eats uses a crowdsourced driver network for rapid food delivery. - Instacart partners with local stores for same-day grocery deliveries. - UPS My Choice allows users to set specific delivery preferences like time windows and drop locations. - Nordstrom uses data and AI to offer tailored delivery options and notifications.	High	Medium	V,high
Developing New Revenue streams 	- Innovative marketing on vehicles/fleet for other companies can act as additional source of revenue. - Monetization of data and logistical infrastructure can be another source - New services like packaging, warehousing, 3rd party tracking services can be provided - Handling of import/export documentation cost while last mile delivery - Freight Insurance services with partnered provider - Cross selling packing materials	UPS: ad space on its trucks. FedEx: data insights and monetization Amazon: warehouse space is rented when not in use DHL: warehousing, packaging services UPS: freight insurance service Customhouse Brokers: import/export paperwork handling Uline: cross sells packaging materials	High	Medium	V.High

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